

# LAB 02: The Alarm That Never Fired

Diagnosing silent failures, understanding CloudWatch states, and building trustworthy observability.

NotebookLM

SEV-3

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We added an **alarm** after the outage — it never fired during our failover test.

## The Stakes

A safety net that doesn't catch you is worse than no safety net at all. It provides a false sense of security.

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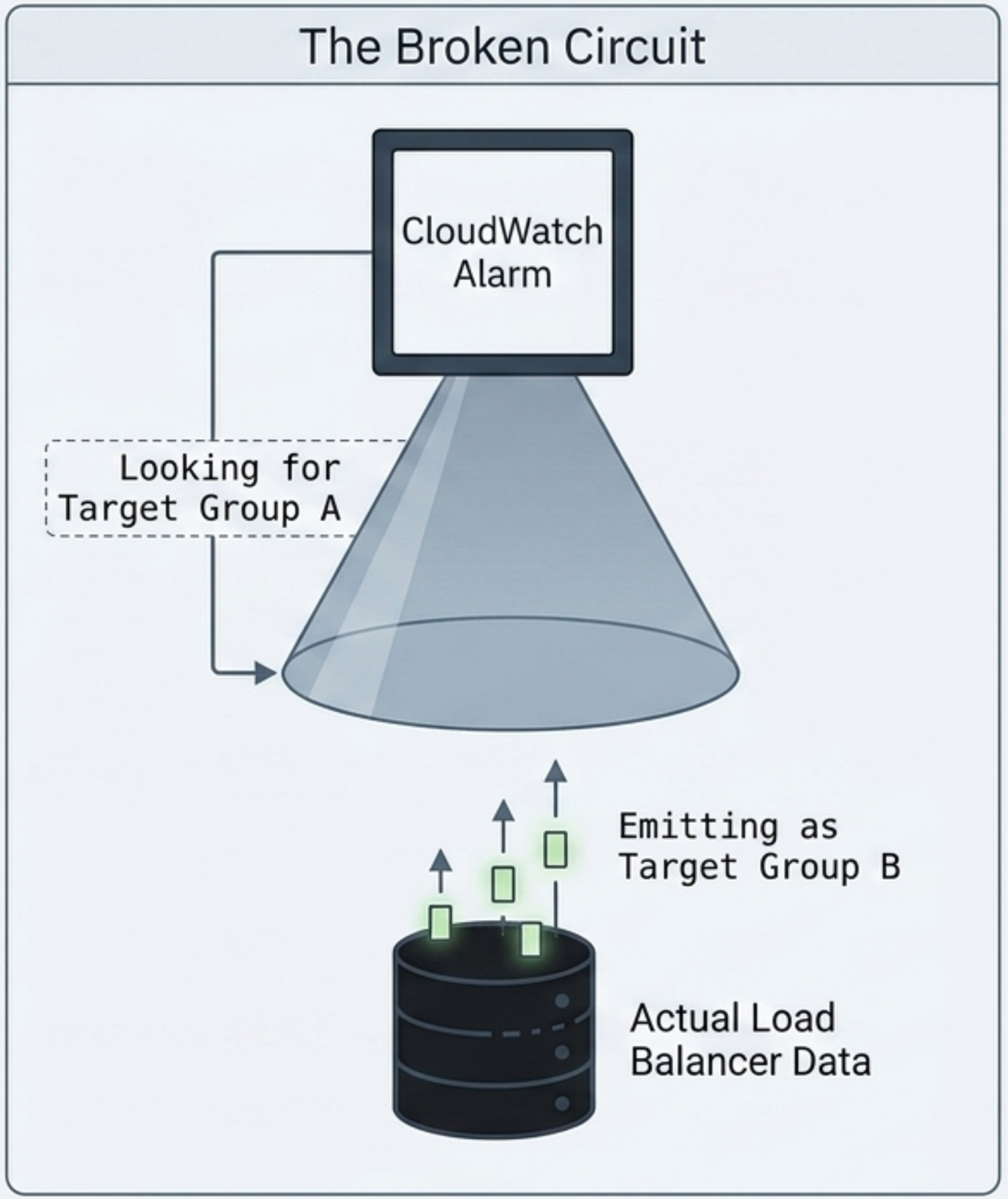
| State | Indicator | Meaning | Operational Reality |
| --- | --- | --- | --- |
| `OK` | ✅ | Metric is within defined thresholds. | System is healthy. |
| `ALARM` | 🔔 | Metric has breached thresholds. | Action required. SNS fires. |
| `INSUFFICIENT_DATA` | ❓ | Not enough data to determine state. | **Danger.** Often mistaken for fine. The alarm is blind. |

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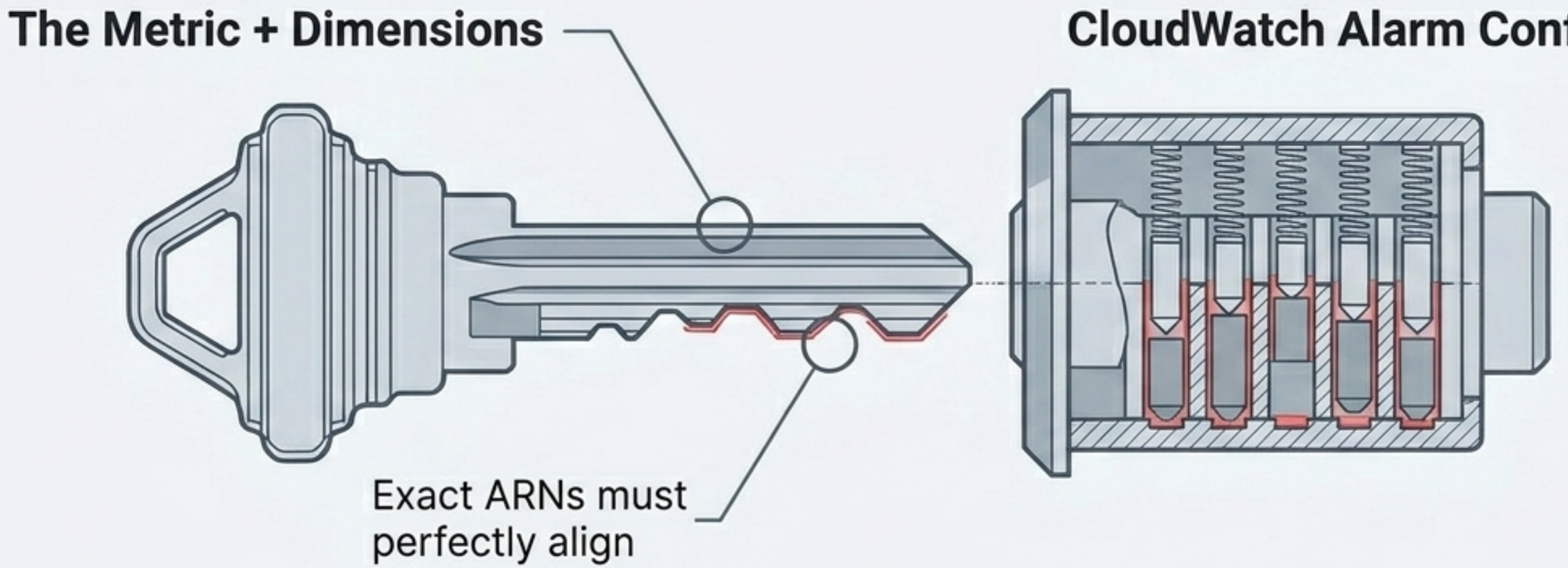
```
$ export ALB_ARN, TG_ARN from terraform output
$ ./break.sh
$ describe-alarms

    State: INSUFFICIENT_DATA

    Notice: This is NOT an OK state.
```




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## The Pathology

An alarm is only as good as its precise metric and dimensions. The `break script` pointed the alarm at the wrong target group. If dimensions are off by even a single character, the key doesn't fit the lock. CloudWatch sees nothing. Blindness equals `INSUFFICIENT_DATA`.

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# The Remediation Pipeline

## 1. Destination

```
$ sns create-topic
+ subscribe
```



Create a real place for the alarm to notify.

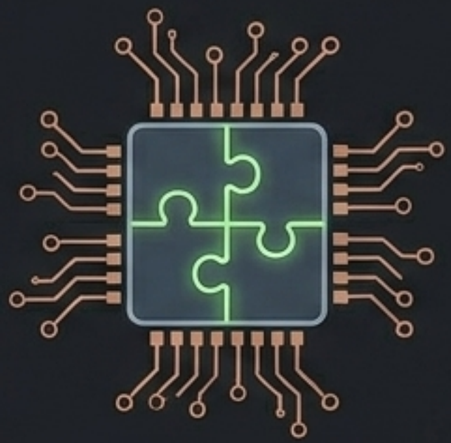
## 2. Precision

```
$ put-metric-alarm
```



Rebuild the alarm using exact, correct target group dimensions.

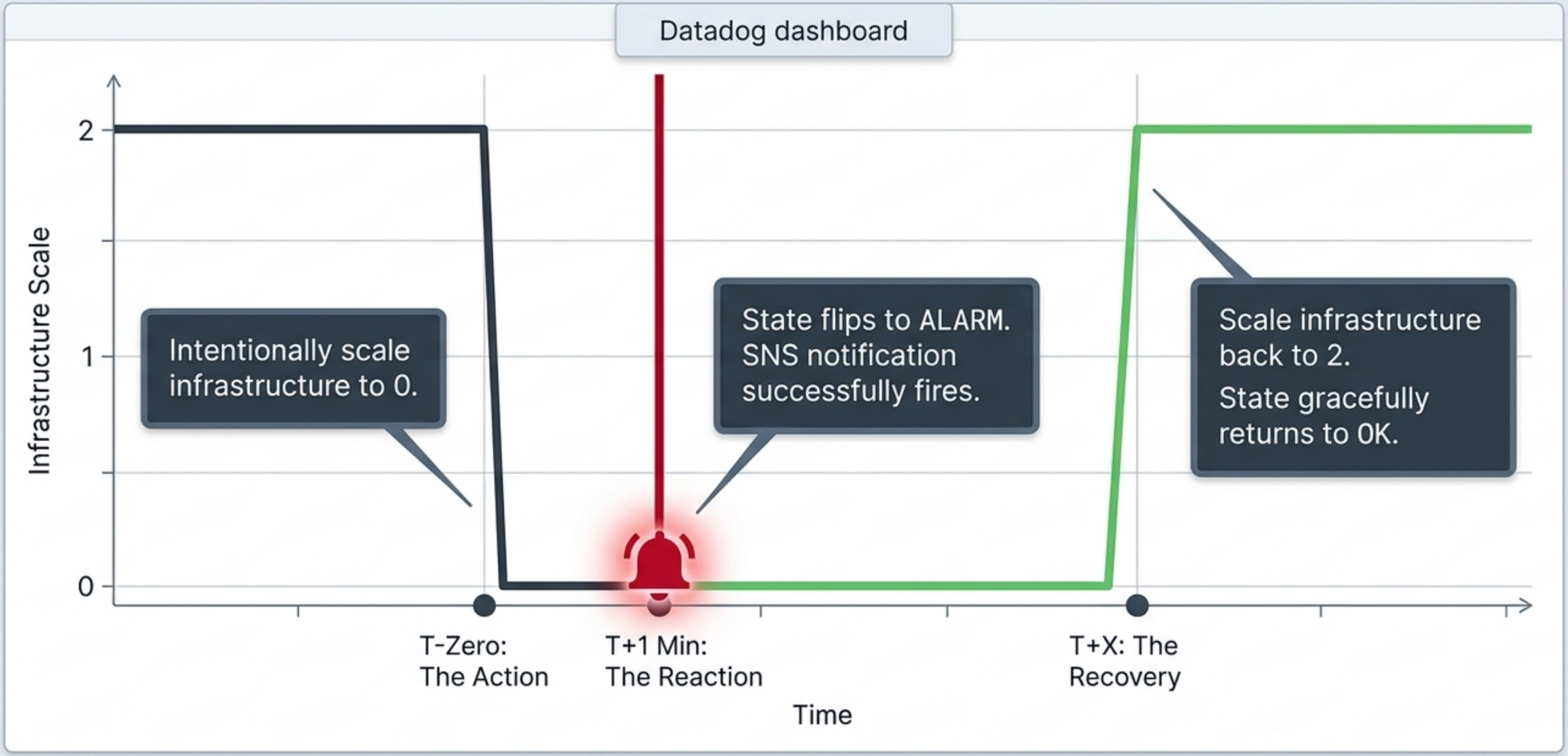
## 3. Connection



Wire the CloudWatch alarm action directly to the SNS topic ARN.

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# Incident Timeline: Intentional Scale & Recovery




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## Untested = Unreliable

An untested alarm is merely a guess, not a safety net.

Validation via intentional failure (chaos testing) is a mandatory Day-2 operation.

You don’t have an alarm until you’ve watched it fire.



## Grey is Not Green

`INSUFFICIENT_DATA` is a critical red flag, never a fine state.

It requires immediate investigation into metric dimensions, ARN alignment, and infrastructure health.

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